

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

QUIZ 2 SET B**SUMMER SEMESTER, 2023-2024****DURATION: 30 MINUTES****FULL MARKS: 20**

CSE 4205: Digital Logic Design

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer **all 3 (three)** questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

ID:

1. Use Karnaugh map to find the minimal form for the following expression. Clearly group the terms contributing to your solution.

5
(CO3)
(PO2)

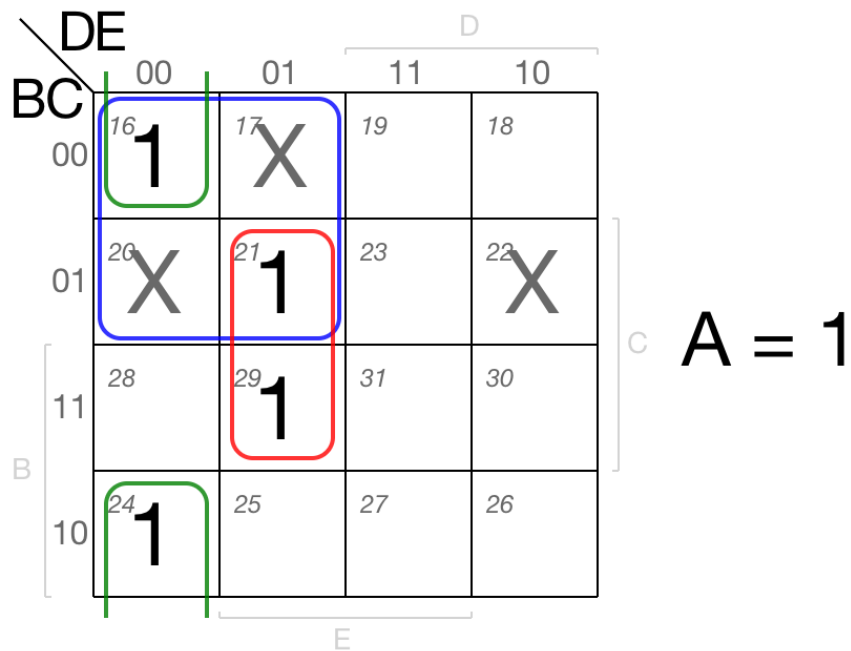
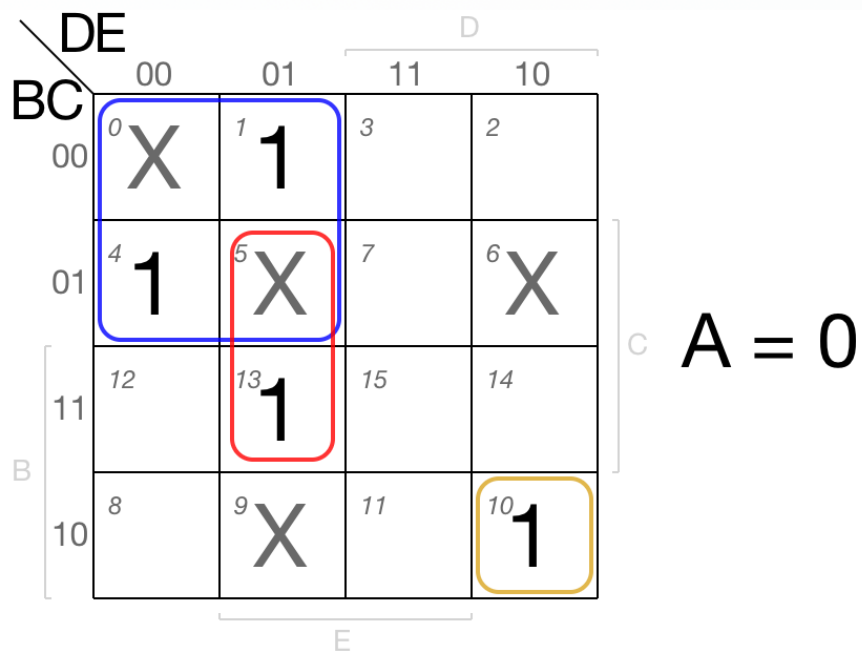
$$F(A, B, C, D, E) = \sum m(1, 4, 10, 13, 16, 21, 24, 29) + \sum d(0, 5, 6, 9, 17, 20, 22)$$

Solution:**Final Answer:**

$$F = B'D' + CD'E + AC'D'E' + A'BC'DE'$$

Rubrics:

- 0.5 point each for correctly grouping each term (times 5).
- 0.5 point each for correctly identifying each term (times 5).
- No point for drawing the K-map since its already given.



$$F = B'D' + CD'E + AC'D'E' + A'BC'DE'$$

Figure 1: K-map solution

2. Simplify the following expression using Quine McCluskey (Tabular Method). And in the process identify the Prime Implicants and Essential Prime Implicants.

11
(CO5)
(PO3)

$$F(A, B, C, D) = \sum m(1, 3, 7, 9, 10) + \sum d(5, 12)$$

Solution:

For detailed solution visit this site. **Final Answer:**

$$F = AB'CD' + B'C'D + A'D$$

Rubrics:

- 1 point for correctly writing the binary representation of each minterm.
- 1 point for correctly grouping minterms based on number of 1s in their binary representation.
- 3 points for correct matching table 1.
- 3 points for correct matching table 2.
- 2 points for correct Prime Implicant Table.
- 1 point for correctly identifying Essential Prime Implicants.

3. Simplify the following Boolean expression using the Cut and Try method, and draw the simplified Boolean circuit using only NOR gates.

4
(CO2)
(PO2)

$$F = xy'z + x'z + xy$$

Solution:

Given,

$$\begin{aligned} F &= x\bar{y}z + \bar{x}z + xy \\ \Rightarrow F &= x\bar{y}z + xy + \bar{x}z \\ \Rightarrow F &= x(\bar{y}z + y) + \bar{x}z \\ \Rightarrow F &= x(\bar{y} + y)(z + y) + \bar{x}z \\ \Rightarrow F &= x(z + y) + \bar{x}z \\ \Rightarrow F &= xz + xy + \bar{x}z \\ \Rightarrow F &= z(x + \bar{x}) + xy \\ F &= xy + z \end{aligned}$$

NOR Representation -

$$F = z + \overline{\overline{x} + \overline{y}}$$

Try to draw it on your own.

Final Answer:

$$F = xy + z$$

$$F = z + \overline{\overline{x} + \overline{y}}$$

Rubrics:

- 2 points for correct simplification.
- 2 points for correct NOR representation.